

Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```
library(ggplot2)
```

```
library(gridExtra)
```

```
library(reshape2)
```

Product Sales Dataset

```
sales <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

Convert data to long format

```
sales_long <- melt(sales, id.vars = "Product",  
  variable.name = "Month",  
  value.name = "Sales")
```

1. Grouped Bar Chart

```
p1 <- ggplot(sales_long,  
  aes(x = Month, y = Sales, fill = Product)) +  
  geom_bar(stat = "identity", position = "dodge") +
```

```
labs(  
  title = "Monthly Product Sales",  
  x = "Month",  
  y = "Sales"  
)  
print(p1)
```

2. Stacked Area Chart

```
p2 <- ggplot(sales_long,  
  aes(x = Month,  
    y = Sales,  
    fill = Product,  
    group = Product)) +  
  geom_area(alpha = 0.7) +  
  labs(  
    title = "Overall Sales Trend",  
    x = "Month",  
    y = "Sales"  
  )  
print(p2)
```

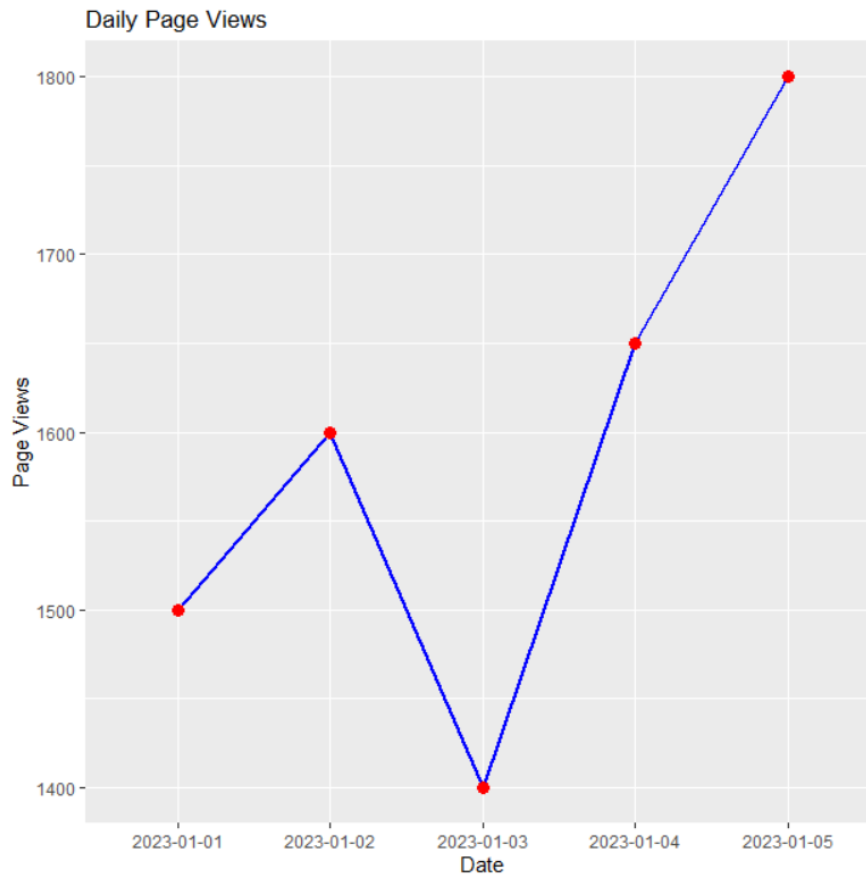
3. Monthly Sales Table

```
print("Monthly Product Sales Data")  
print(sales)
```

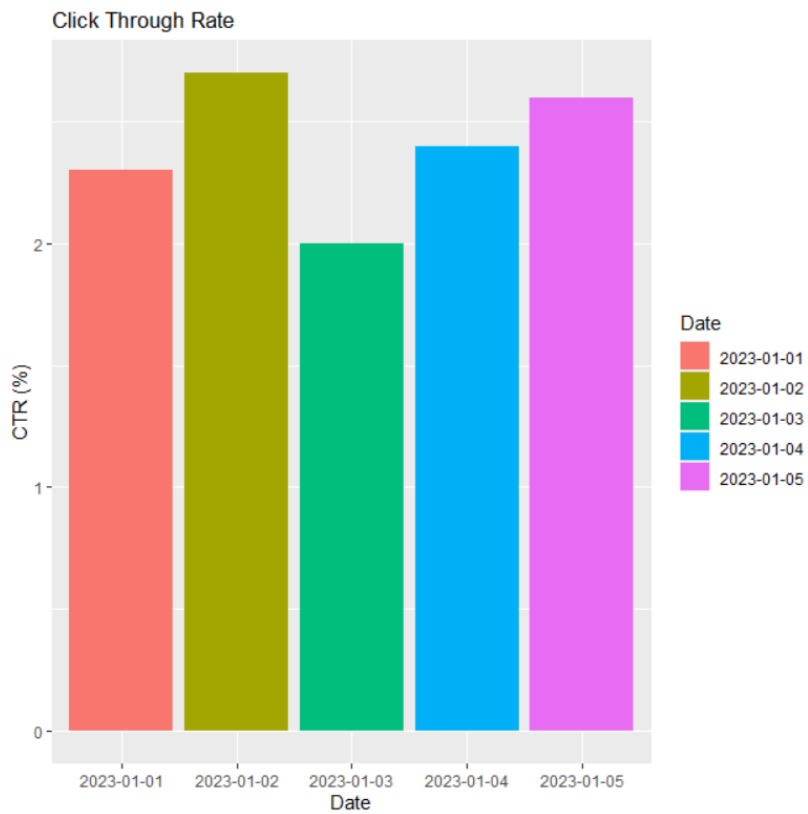
4. Dashboard

```
grid.arrange(p1, p2, ncol = 2)
```

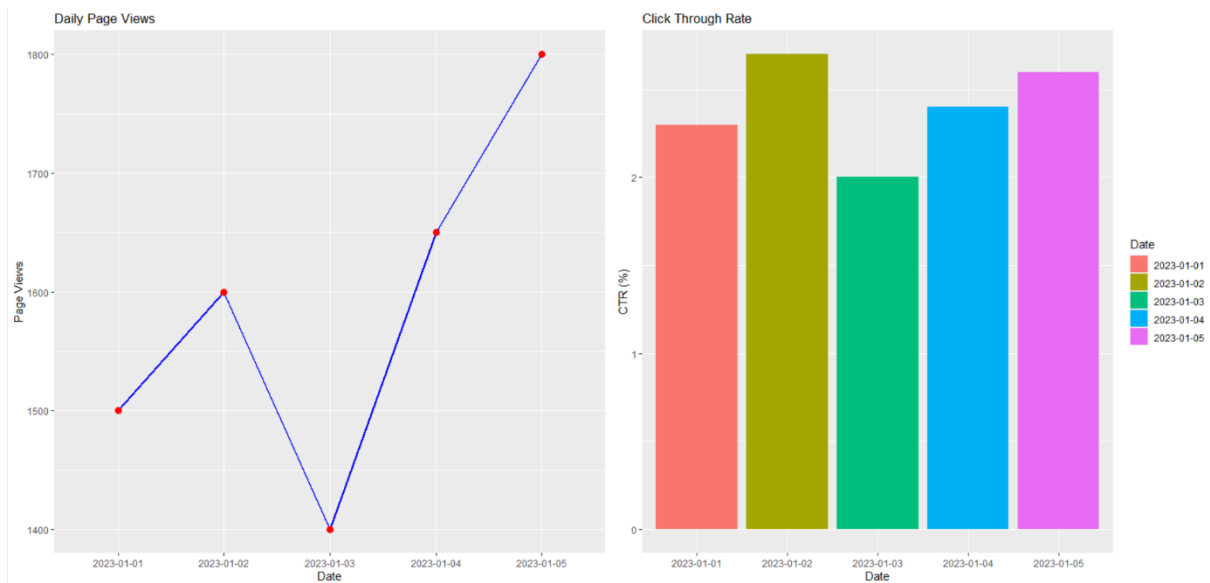
1. Grouped Bar Chart



2. Stacked Area Chart



4. Dashboard



Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

```
# Load Libraries
```

```
library(ggplot2)
```

```
library(gridExtra)
```

```
# Customer Demographics Dataset
```

```
customer <- data.frame(
```

```
  CustomerID = c(1, 2, 3),
```

```
  Age = c(28, 35, 42),
```

```
  Gender = c("Female", "Male", "Female"),
```

```
  Income = c(50000, 60000, 75000)
```

```
)
```

1. Bar Chart - Customer Age Distribution

```
p1 <- ggplot(customer, aes(x = factor(CustomerID), y = Age, fill = factor(CustomerID))) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Customer Age Distribution",  
    x = "Customer ID",  
    y = "Age"  
  )  
  
print(p1)
```

2. Pie Chart - Gender Distribution

```
gender_count <- table(customer$Gender)  
  
pie(  
  gender_count,  
  labels = paste(names(gender_count), "(", gender_count, ")", sep = ""),  
  main = "Customer Gender Distribution",  
  col = c("lightblue", "pink")  
)
```

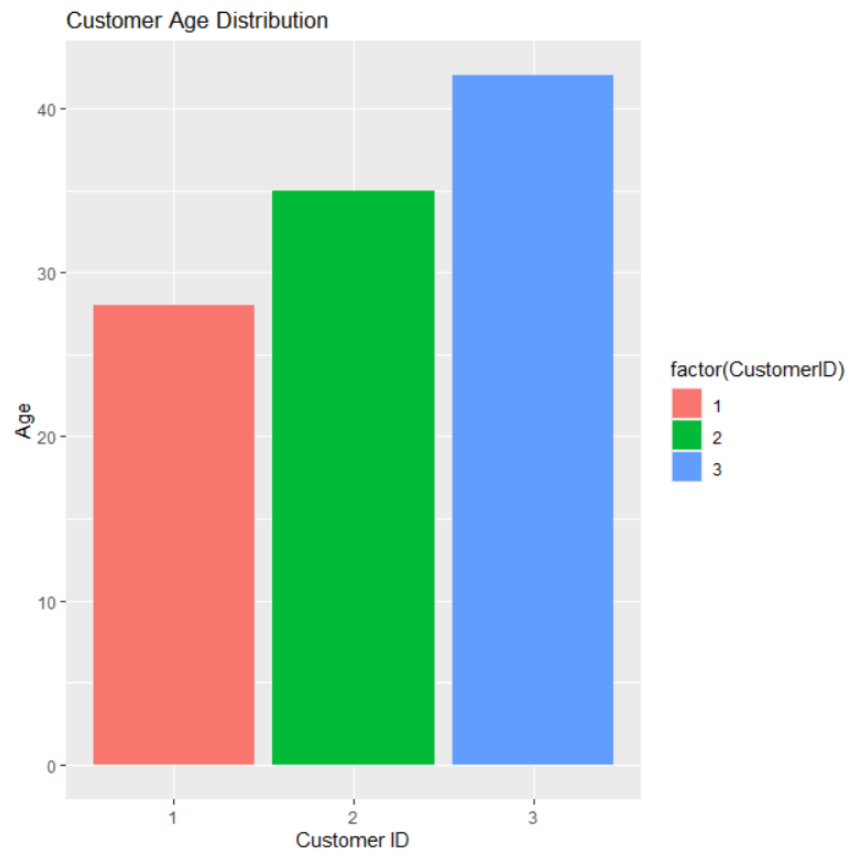
3. Customer Demographics Table

```
print("Customer Demographics Data")  
print(customer)
```

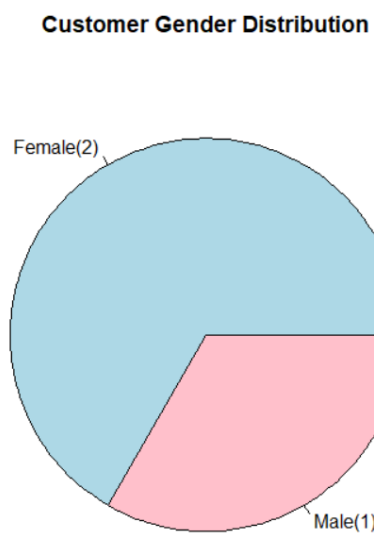
4. Dashboard

```
grid.arrange(p1, ncol = 1)
```

1. Bar Chart - Customer Age Distribution



2. Pie Chart - Gender Distribution



Set 8 Employee Performance Analysis
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85

2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Load Libraries

```
library(ggplot2)
```

```
library(gridExtra)
```

Employee Performance Dataset

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  YearsService = c(5, 3, 7),  
  PerformanceScore = c(85, 92, 78)  
)
```

1. Line Chart - Performance Trend

```
p1 <- ggplot(employee, aes(x = YearsService, y = PerformanceScore, group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +
```

```
geom_point(color = "red", size = 3) +  
labs(  
  title = "Employee Performance Trend",  
  x = "Years of Service",  
  y = "Performance Score"  
)
```

```
print(p1)
```

2. Bar Chart - Department-wise Performance

```
p2 <- ggplot(employee, aes(x = Department, y = PerformanceScore, fill = Department)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Employee Performance by Department",  
    x = "Department",  
    y = "Performance Score"  
  )
```

```
print(p2)
```

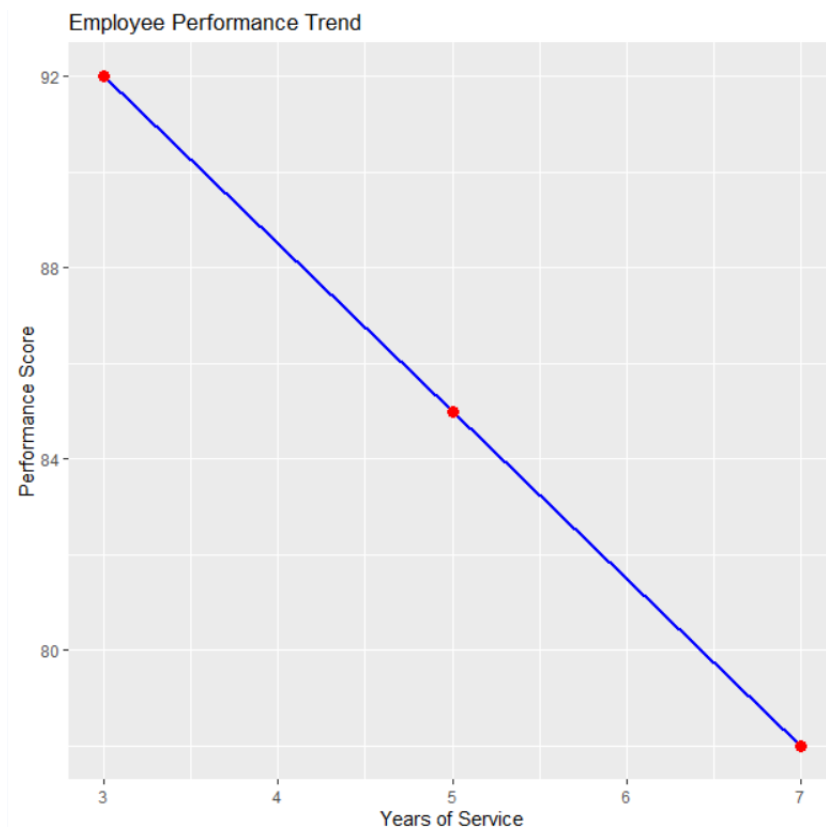
3. Employee Performance Table

```
print("Employee Performance Data")  
print(employee)
```

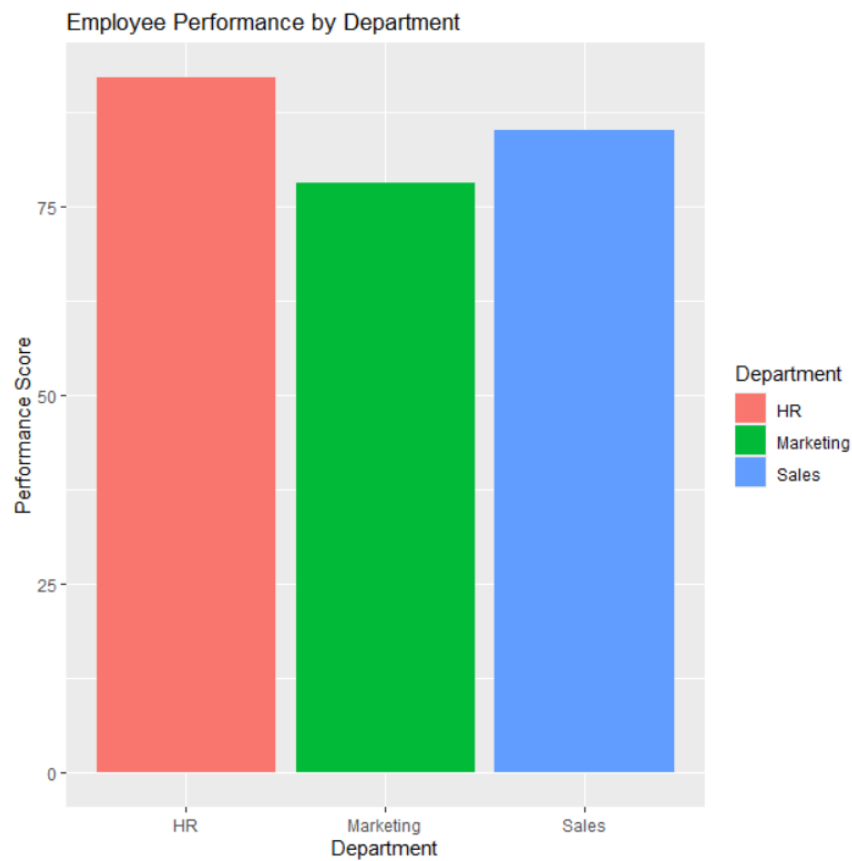
4. Dashboard

```
grid.arrange(p1, p2, ncol = 2)
```

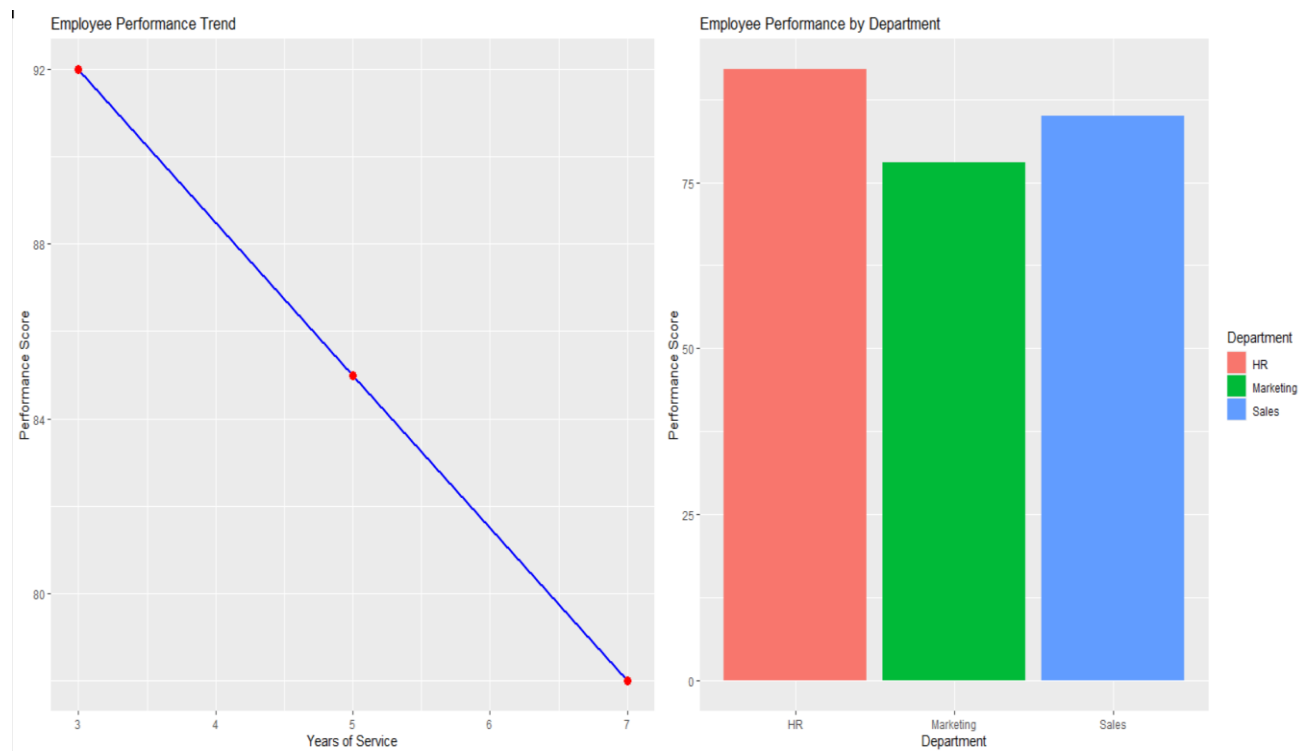

1. Line Chart - Performance Trend



2. Bar Chart - Department-wise Performance



4. Dashboard



Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85

DEA

2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

```
library(ggplot2)
library(gridExtra)
```

Employee Performance Dataset

```
employee <- data.frame(
  EmployeeID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  YearsService = c(5, 3, 7),
  PerformanceScore = c(85, 92, 78)
)
```

1. Line Chart - Employee Performance Trend

```
p1 <- ggplot(employee, aes(x = YearsService, y = PerformanceScore, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()

print(p1)
```

2. Bar Chart - Employees by Department

```
p2 <- ggplot(employee, aes(x = Department, y = PerformanceScore, fill =
Department)) +
  geom_bar(stat = "identity") +
  labs(
```

```

        title = "Performance Score by Department",
        x = "Department",
        y = "Performance Score"
    ) +
    theme_minimal()

print(p2)

# 3. Employee Performance Table

cat("\nEmployee Performance Data\n")
print(employee)

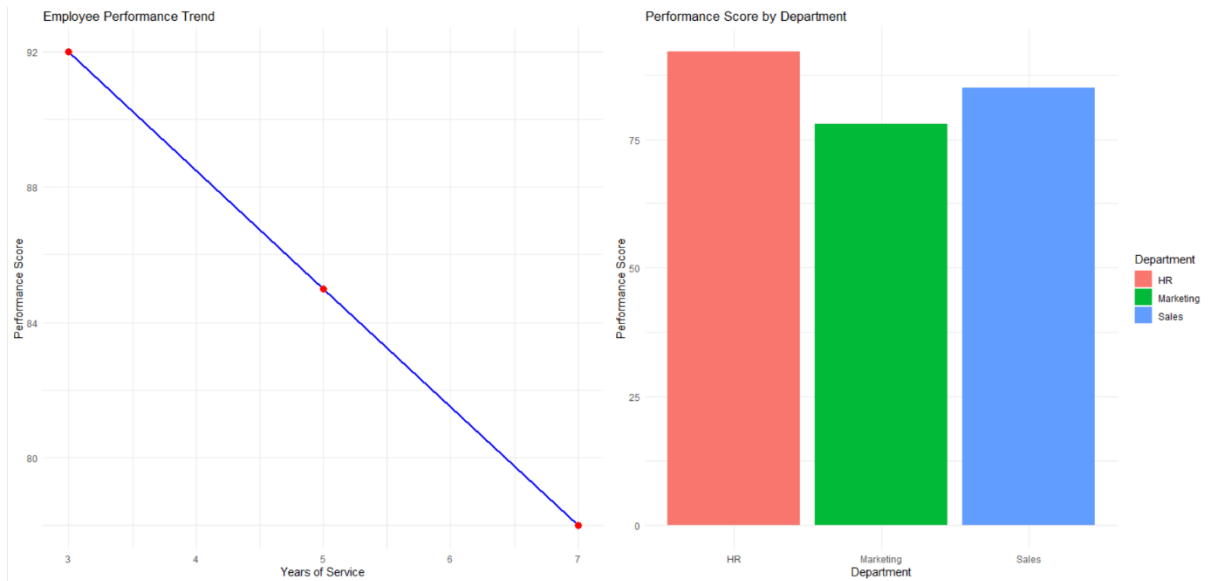
# Save dataset for Tableau
write.csv(employee, "Employee_Performance.csv", row.names = FALSE)

# 4. Dashboard

grid.arrange(p1, p2, ncol = 2)

cat("\nEmployee_Performance.csv has been created successfully.\n")
cat("Import this CSV file into Tableau to create the dashboard.\n")

```



set 9

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

1. Import the dataset in **R**. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.
2. Create a scatter plot **using R programming** between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.
3. Using **R programming**, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.
4. Import Online_Learning_Activity.csv **into Tableau**. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

```
library(ggplot2)
```

```
# Dataset
```

```
online <- data.frame(
```

```
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
```

```
  Gender = c("Male","Female","Male","Female","Male","Female"),
```

```
Age = c(20,22,19,21,23,20),  
Course = c("R","R","SQL","R","R","SQL"),  
Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
Videos_Watched = c(12,15,8,18,9,14),  
Quiz_Score = c(78,85,65,92,70,88),  
Login_Date = c("2025-01-05","2025-01-05","2025-02-08",  
               "2025-02-08","2025-03-12","2025-03-12")  
)
```

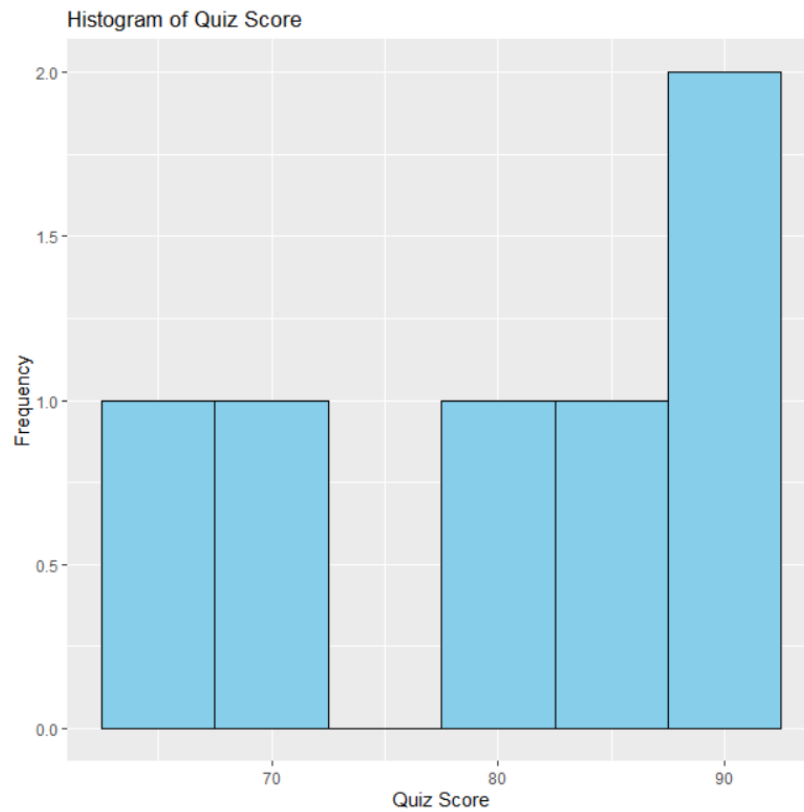
```
# Histogram
```

```
ggplot(online, aes(x = Quiz_Score)) +  
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +  
  labs(title = "Histogram of Quiz Score",  
        x = "Quiz Score",  
        y = "Frequency")
```

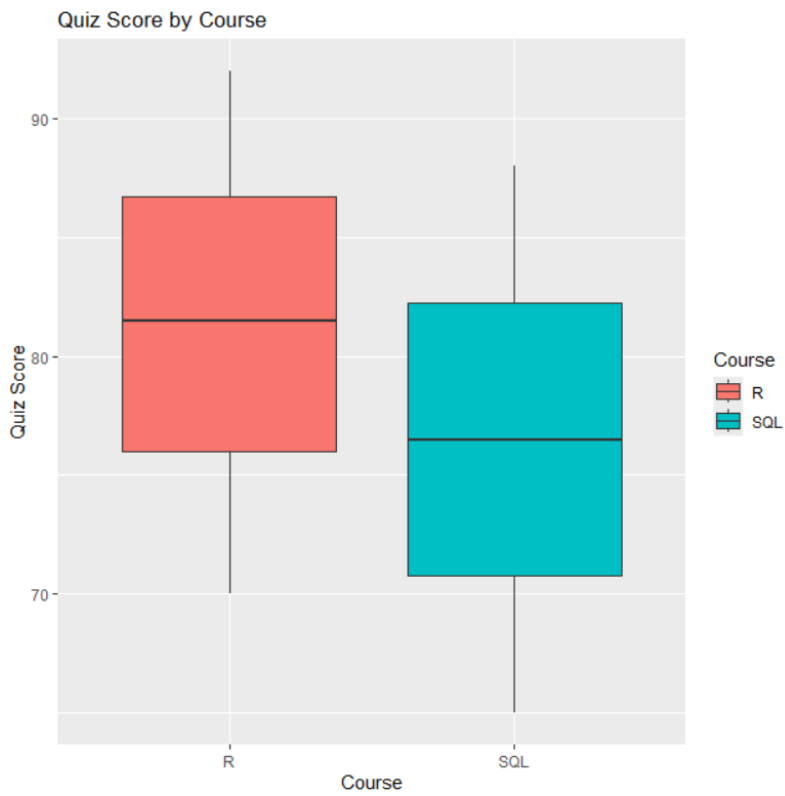
```
# Boxplot
```

```
ggplot(online, aes(x = Course, y = Quiz_Score, fill = Course)) +  
  geom_boxplot() +  
  labs(title = "Quiz Score by Course",  
        x = "Course",  
        y = "Quiz Score")
```

Histogram



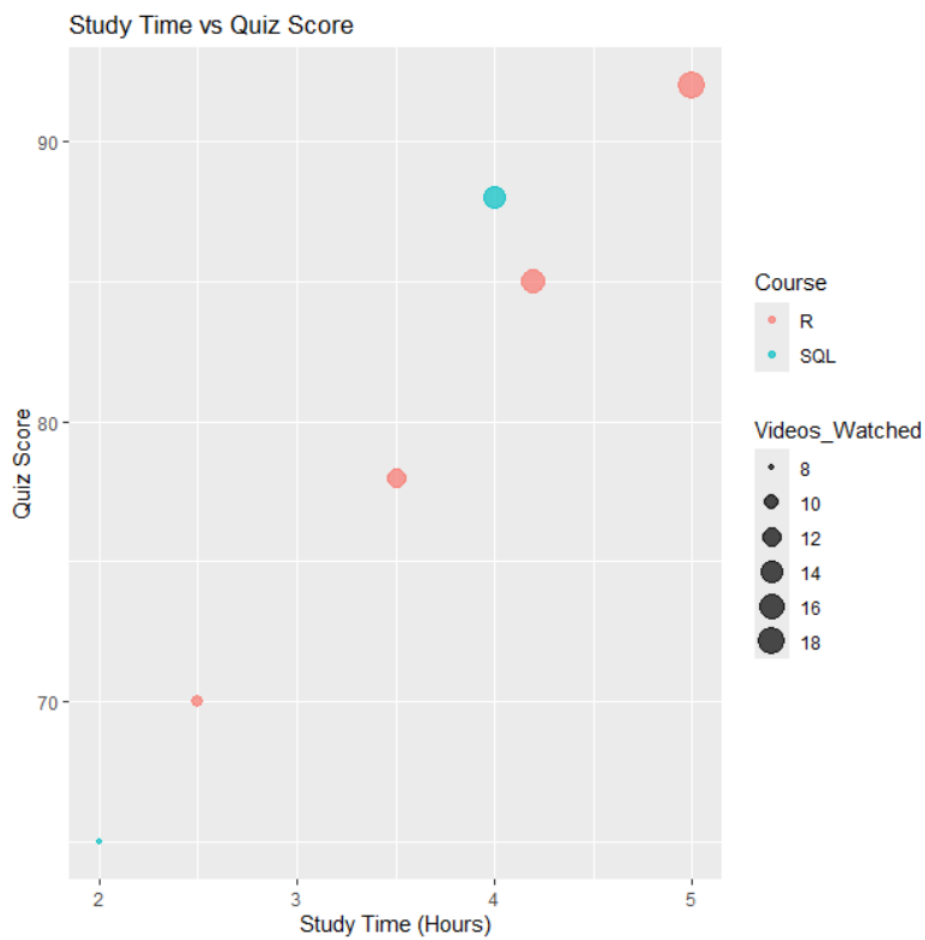
Boxplot



2.Scatter Plot

```
library(ggplot2)
```

```
ggplot(online, aes(x = Study_Time,  
                  y = Quiz_Score,  
                  size = Videos_Watched,  
                  color = Course)) +  
  geom_point(alpha = 0.7) +  
  labs(title = "Study Time vs Quiz Score",  
        x = "Study Time (Hours)",  
        y = "Quiz Score")
```



3.Line Chart

```
library(dplyr)
```

```
# Convert Date
```

```
online$Login_Date <- as.Date(online$Login_Date)
```

```
# Monthly Average
```

```
monthly <- online %>%
```

```
  mutate(Month = format(Login_Date,"%Y-%m")) %>%
```

```
  group_by(Month) %>%
```

```
  summarise(Average_Score = mean(Quiz_Score))
```

```
# Line Chart
```

```
ggplot(monthly,
```

```
  aes(x = Month,
```

```
      y = Average_Score,
```

```
      group = 1)) +
```

```
geom_line(color = "blue") +
```

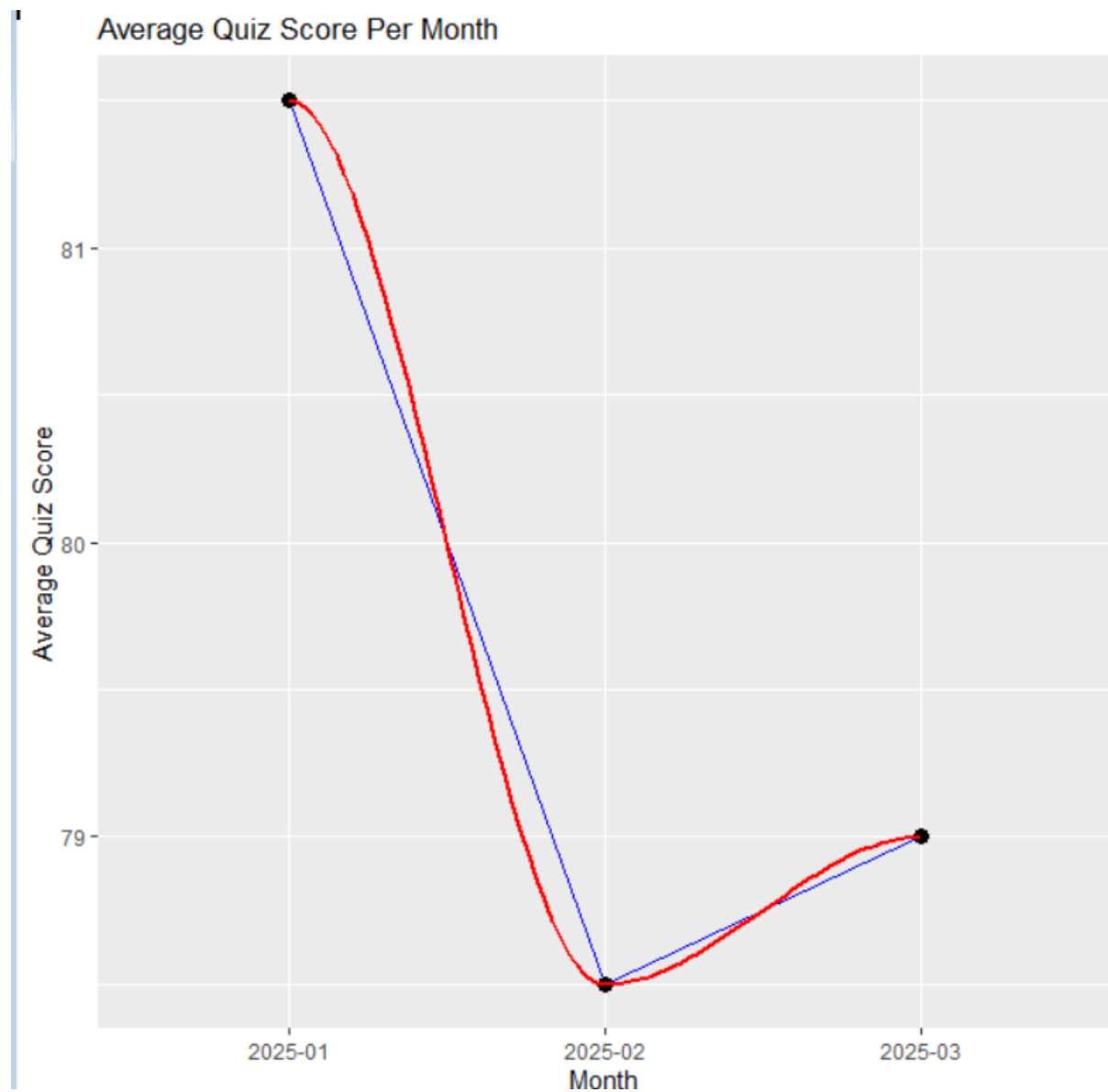
```
geom_point(size = 3) +
```

```
geom_smooth(method = "loess", se = FALSE, color = "red") +
```

```
labs(title = "Average Quiz Score Per Month",
```

```
      x = "Month",
```

```
      y = "Average Quiz Score")
```



Tableau

```
write.csv(online,  
          "Online_Learning_Activity.csv",  
          row.names = FALSE)
```

Display Dataset

```
print(online)
```

Set 10 Survey Responses Analysis
Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart

ALMAS BEGUM posted a new assignment: Class Activity

elements.

2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

1: Grouped Bar Chart

Load Libraries

```
library(ggplot2)
```

```
library(gridExtra)
```

```
library(reshape2)
```

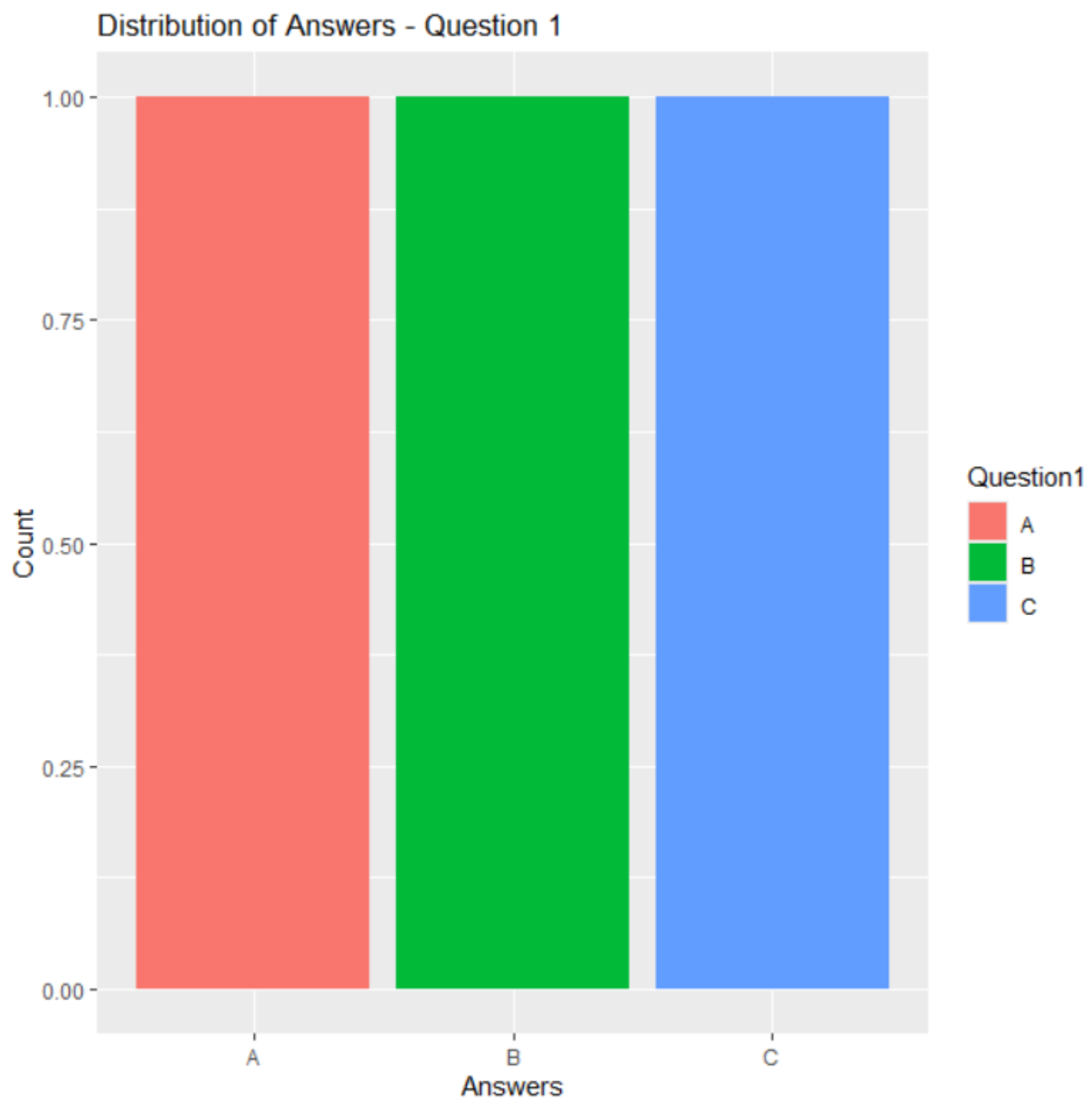
Survey Dataset

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
# Grouped Bar Chart for Question 1

p1 <- ggplot(survey, aes(x = Question1, fill = Question1)) +
  geom_bar() +
  labs(
    title = "Distribution of Answers - Question 1",
    x = "Answers",
    y = "Count"
  )

print(p1)
```



2: Stacked Bar Chart

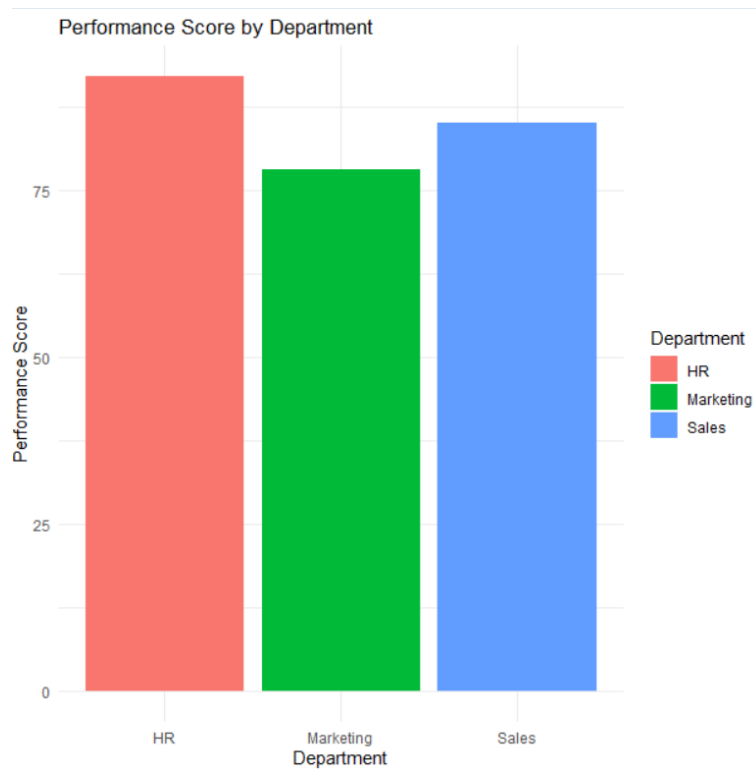
Convert data to long format

```
survey_long <- melt(survey, id.vars = "SurveyID")
```

Stacked Bar Chart

```
p2 <- ggplot(survey_long,  
             aes(x = variable, fill = value)) +  
  geom_bar() +  
  labs(  
    title = "Overall Survey Response Distribution",  
    x = "Questions",  
    y = "Count"  
  )
```

```
print(p2)
```



```
# Display Survey Data
```

```
print(survey)
```

```
# Save Dataset for Tableau
```

```
write.csv(survey, "Survey_Responses.csv", row.names = FALSE)
```

```
# Dashboard
```

```
grid.arrange(p1, p2, ncol = 2)
```

